

KILL DEVIL HILLS, NC, USA

WMO: 720282

Lat: 36.018N

Lon: 75.671W

Elev: 13

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.1	29.5	5.4	7.3	27.6	11.6	10.0	34.5	19.9	40.9	18.3	40.8	9.0	310	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	92.4	78.2	90.0	77.3	87.6	76.2	80.8	88.2	79.2	86.3	78.0	84.3	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.7	84.7	77.3	142.1	82.5	76.1	136.6	81.6	44.2	87.9	42.6	86.4	41.4	84.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.7	13.0	21.6	97.2	4.9	3.0	18.1	99.3	15.3	101.0	12.5	102.7	9.0	104.9		
			DB	WB	18.4	82.2	5.0	2.3	14.8	83.8	11.9	85.1	9.0	86.4	5.4	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	44.4	44.8	50.6	59.8	68.3	76.8	80.2	79.4	75.4	66.4	55.3	49.4
	DBStd	14.82	9.26	8.46	9.09	7.77	6.91	5.23	4.01	4.12	4.56	7.15	7.67	8.58
	HDD50	682	223	188	109	10	0	0	0	0	0	1	34	117
	HDD65	2756	637	566	453	193	49	1	0	0	1	71	300	485
	CDD50	5308	51	42	128	304	568	804	937	912	761	509	193	99
	CDD65	1907	0	0	7	37	152	355	472	447	311	114	9	3
	CDH74	14492	0	1	24	155	788	2801	4539	3935	1828	404	14	3
	CDH80	4162	0	0	1	18	159	856	1520	1204	359	44	1	0
Wind	WSAvg	6.0	6.8	7.0	7.0	6.8	6.2	5.5	5.0	4.5	5.9	5.7	6.1	5.9
Precipitation	PrecAvg	55.3	4.3	3.7	4.4	3.7	4.3	5.0	6.2	6.0	5.7	4.2	3.5	4.2
	PrecMax	73.7	7.5	7.8	8.8	7.0	8.0	9.4	10.0	11.5	19.9	8.7	9.3	8.6
	PrecMin	38.5	1.8	1.4	1.3	1.3	1.7	1.8	3.7	2.4	1.4	0.1	0.8	1.7
	PrecStd	7.4	1.4	1.6	1.5	1.4	1.7	1.9	1.5	2.1	3.9	1.9	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	70.6	77.3	82.3	87.7	95.0	96.0	96.8	90.3	84.1	75.4	72.3
		MCWB	61.3	60.9	63.7	66.2	71.7	76.7	79.8	82.2	77.2	73.1	67.4	67.3
	2%	DB	65.0	66.5	72.5	78.4	84.2	90.6	92.6	91.2	86.3	80.9	71.8	68.4
		MCWB	61.0	58.7	60.9	65.6	71.4	75.8	78.6	79.0	75.7	72.9	65.2	64.8
	5%	DB	62.5	62.8	68.3	74.9	81.2	87.7	90.1	88.5	83.8	78.4	68.5	65.3
		MCWB	58.0	56.8	60.1	64.6	69.9	74.9	77.6	77.7	74.8	71.4	63.8	62.3
	10%	DB	58.7	58.5	64.3	71.7	78.5	84.9	87.5	86.1	81.7	76.0	66.1	62.7
		MCWB	54.8	53.3	58.7	63.4	69.1	74.0	76.5	76.8	73.9	70.5	62.6	59.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.7	64.0	66.3	71.1	75.1	79.6	82.2	84.0	80.2	76.5	69.8	68.9
		MCDB	65.9	68.2	73.5	76.1	81.4	88.3	91.3	93.2	85.3	79.9	72.6	70.7
	2%	WB	61.8	61.0	64.0	68.6	73.4	77.9	80.3	81.7	78.0	74.9	67.1	65.6
		MCDB	65.0	64.6	69.3	74.6	80.2	86.1	89.4	88.4	82.7	78.5	69.8	68.1
	5%	WB	58.8	57.4	62.0	66.8	72.2	76.8	78.9	79.7	76.9	73.4	64.9	62.8
		MCDB	62.0	61.9	66.7	72.6	78.4	84.6	87.3	85.2	81.4	77.3	67.6	64.8
	10%	WB	55.1	53.4	59.5	64.8	70.9	75.5	78.0	78.0	75.8	71.8	62.6	59.9
		MCDB	58.2	57.7	64.2	70.6	76.7	82.7	85.0	83.5	80.4	75.6	65.6	62.3
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.1	13.4	14.1	12.9	12.2	11.9	10.8	9.3	10.6	11.1	11.6
		MCDBR	17.0	20.2	19.6	19.1	16.9	15.8	15.5	14.3	12.3	12.5	13.8	14.8
		MCWBR	14.2	14.8	12.0	9.9	7.9	6.7	6.6	6.0	6.0	6.5	10.2	12.5
	5% WB	MCDBR	16.7	19.5	18.0	17.2	14.4	14.4	14.5	14.0	10.9	10.5	12.3	14.4
MCWBR		15.4	16.6	13.2	10.6	7.5	6.6	6.9	6.4	6.3	6.4	11.0	13.2	
Clear-Sky Solar Irradiance	taub	0.323	0.331	0.354	0.398	0.441	0.486	0.513	0.501	0.418	0.368	0.336	0.328	
	taud	2.536	2.510	2.469	2.356	2.274	2.199	2.131	2.154	2.388	2.512	2.546	2.549	
	Ebn at Noon	278	289	290	281	269	256	248	248	266	272	271	269	
	Edh at Noon	25	29	33	39	43	46	49	47	36	29	25	24	
All-Sky Solar Radiation	RadAvg	749	998	1342	1714	1928	2043	1928	1754	1452	1163	847	677	
	RadStd	53	100	101	123	139	111	97	117	94	114	61	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+196

Nomenclature: See separate page